Large Language Models for Bayesian Network Parameterization in Software Engineering: Direct Elicitation versus Ranked Nodes

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Abstract

Software engineering (SE) decision makers need models they can understand and use when data are scarce. Bayesian networks (BNs) make probabilistic relationships explicit and incorporate expert knowledge, but eliciting their conditional probability tables (CPTs) is costly. The Ranked Nodes Method (RNM) generates ordinal CPTs from fewer judgments. Should large language models (LLMs) fill complete CPTs or supply fewer judgments to RNM? We compared complete CPTs for 61 ordinal conditional nodes from seven SE BNs in an exploratory study. We tested structured prompts that clarified conditioning, framed probabilities as hypothetical frequencies, and requested consistency checks. Qualitative retrieval-augmented generation (RAG) supplied short passages on domain relationships, without reference probabilities or numerical parameters. Eight conditions used GPT-6 Astra at medium effort across five replications. With equal node weights, total variation averaged 0.2317 for direct elicitation and 0.2404 for RNM, a 3.76% increase. Because nodes were unevenly distributed across five source studies, we weighted each study equally; direct elicitation still had lower error. Both contrasts passed Holm-adjusted paired t-tests, but not exact sign tests. RNM improved 31 nodes, yet larger deteriorations elsewhere outweighed those gains. Structured prompting modestly helped direct elicitation under node weighting; qualitative RAG demonstrated no benefit. RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, but only 0.61% fewer total tokens. The findings favor direct elicitation as the average numerical baseline in this sample and show why reduced elicitation alone cannot establish a better, cheaper to complete CPT.

CCS Concepts

• **Computing methodologies** → **Bayesian networks**; • **Software and its engineering** → *Software creation and management*.

Keywords

Bayesian networks, large language models, probability elicitation, ranked nodes, software engineering

1 Introduction

Supporting software engineering (SE) decisions faces two main challenges: explaining the basis for a recommendation and obtaining enough relevant data to build a model experts can rely on. Examples include release readiness, quality assessment, and process improvement, where decision makers must judge whether a recommendation's assumptions apply. A review of 108 studies documents the demand for explainability across 23 SE tasks [3]. Training data for machine learning and deep learning may also be scarce, particularly for managerial SE decisions [23]. Explicit assumptions and domain knowledge are therefore valuable.

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Bayesian networks (BNs) offer such an approach. A BN makes relationships and uncertainty explicit in a graph and probabilities, incorporating domain knowledge when data are limited [10, 11]. BNs have supported many SE problems, including early defect prediction [10], effort estimation [17], and diagnosis of development-process problems [21]. A systematic mapping of 106 studies identifies software management and quality as prominent application areas [24]. Building a credible BN nevertheless requires substantial knowledge engineering effort.

Constructing conditional probability tables (CPTs) is a major part of that effort. Each row describes a child's probabilities for one combination of its parents' states; adding parents rapidly increases the number of judgments required [11]. Semi-automatic methods reduce this burden by reconstructing a table from fewer expert judgments or parameters. These include the Ranked Nodes Method (RNM) [11], the Weighted Sum Algorithm (WSA) [20], Analytic Hierarchy Process (AHP)-based pairwise comparisons [5], and interpolation methods [1, 2]. Their assumptions can, however, limit fidelity to the intended probabilities [2].

Large language models (LLMs) are now being explored for CPT construction. Nafar et al. elicited numerical BN parameters directly [18]; Gottal and Matthes reconstructed CPTs from statistical summaries extracted from scientific literature [12]; and Cypko et al. used retrieval-augmented generation (RAG) to support expert BN modeling [6]. Should an LLM specify every row, or supply fewer judgments for a semi-automatic method to complete? Any benefit depends on both the judgments and the reconstruction.

We investigated this question using RNM, which applies to variables with ordered states such as Low, Medium, and High [11, 15]. Instead of requesting the entire CPT, we asked the LLM for distributions at selected parent-state combinations, called *anchors*, and for a rule describing how the parents combine. A numerical fitting procedure then adjusted that rule's weights and uncertainty to match the elicited distributions and generated the complete CPT. We compared this approach with direct complete-table elicitation on 61 ordinal conditional nodes from seven SE BNs.

We also examined support for elicitation. Structured prompts made conditioning explicit and asked for probability and consistency checks; qualitative RAG supplied short passages explaining relevant domain relationships. Crossing the two elicitation methods, two prompt designs, and presence or absence of RAG produced eight conditions. We repeated each five times using GPT-6 Astra at medium effort. This study addresses three research questions:

- **RQ1—Reference agreement:** How does LLM-assisted RNM compare with direct elicitation in complete-CPT agreement across the full SE sample?
- **RQ2—Elicitation support:** How do structured prompting and qualitative RAG affect each method's agreement across complete tables?

- **RQ3—Accuracy and effort:** How does RNM trade reference agreement against requested judgments and observed computational effort?

All three RQs include all 61 nodes and every CPT row, including anchors. The seven BNs cover Scrum process assessment [19], Chrome code review [14], ERP data quality [13], software maintenance logs [8], and software strategic indicators [16]. We give nodes equal weight for the main comparison. Because BNs contribute unequal numbers of nodes, we also weight their five source studies equally and report each BN separately. This evaluates the complete tables available for use rather than restricting attention to configurations omitted from RNM prompts.

Section 2 positions the study in related work, Section 3 describes the experiment, and Section 4 reports the results. Section 5 discusses the RQs and implications for SE, Section 6 examines validity threats, and Section 7 concludes the paper.

2 Related Work

We first review semi-automatic techniques that reduce expert CPT elicitation (Section 2.1), then examine LLM approaches and position our comparison (Section 2.2).

2.1 Semi-automatic CPT construction

Semi-automatic techniques generate CPTs from fewer expert judgments. RNM combines weighted parent functions with truncated-normal distributions for ordinal variables [11, 15]: the function describes parent aggregation and the variance controls uncertainty. Alternatives include WSA, AHP-based pairwise comparisons [5], and interpolation from selected distributions, including InterBeta [1]. Each reduces elicitation through assumptions about the completed table.

Blomaard et al. compare RNM, functional interpolation, and InterBeta using known CPTs to optimize parameters [2]. These studies motivate examining both effort and fidelity when LLMs supply judgments: ordinal eligibility alone does not guarantee RNM compatibility.

2.2 LLMs for CPT construction

LLM approaches differ in elicitation granularity and supplied evidence. Nafar et al. found that requesting one child-state probability at a time lowered median BN KL divergence for seven of eight models, with more queries [18]. Their FullDist baseline requests one conditional row; ours requests the whole CPT. Their result leaves open whether eliciting fewer configurations and reconstructing the rest improves agreement. Gottal and Matthes reconstruct from numerical summaries [12], which constrain probabilities more directly than qualitative RAG. Rahul and Chowdhury's persona-panel preprint evaluates effort and internal consistency, without independent reference-CPT accuracy [22].

Other work evaluates modeling or predictive support. BIRD illustrates neural–symbolic integration [7] through LLM judgments, constrained fitting, and BN inference [9]. Clinical RAG assists BN modelers [6]; AutoElicit evaluates prediction after updating elicited parameter priors [4]. These outcomes complement numerical CPT agreement but cannot establish it.

We connect these research directions by comparing direct elicitation with LLM anchors plus RNM fitting across complete CPTs for 61 nodes from seven SE BNs. We examine structured prompting, qualitative RAG, and token usage, with anchor diagnostics separating elicitation from fitting. Experimental fits use LLM judgments without reference access. Table 1 positions our contribution by supplied evidence, evaluation endpoint, and scope.

3 Method

We tested two ways to obtain a complete CPT from an LLM: requesting every row directly, or requesting selected rows and using RNM to complete the table. Each method was crossed with baseline versus structured prompting and with versus without qualitative RAG, giving eight conditions. Every condition covered the same 61 nodes in five replications ($61 \times 8 \times 5 = 2{,}440$ trials). We compared the resulting complete tables against hidden reference CPTs and recorded token usage. This design tests reference agreement (RQ1), elicitation support (RQ2), and the accuracy–effort trade-off (RQ3).

3.1 BNs and reference CPTs

The sample contains seven BNs from five published SE studies (Table 2). Scrum assesses team and process factors affecting sprint and backlog quality. Chrome 40 and 42 concern code review and post-release defects in two software versions. APS concerns ERP data quality for advanced planning and scheduling; the two software-log BNs concern maintenance logs; SESSI concerns software strategic indicators. Versions from the same study form one publication family.

We selected conditional nodes with complete numerical references and interpretable, ordered child and parent states. Binary variables qualified when their meanings supported an order; nominal variables and Boolean gate chains were excluded. The selected CPTs contain 421 rows, with at most 27 rows per node. Scrum uses an RNM-derived reference; the other BNs supply tabulated probabilities. These references represent particular modeling contexts, not universal company probabilities. The supplement describes their versions and preparation.

All three RQs evaluate complete CPTs, including RNM anchors and remaining rows. Nodes with one parent or with all rows elicited are retained because fitting can still change their probabilities. The sample is availability-based, and the complete-table endpoint, weighting, and tests were adopted after inspecting earlier results; the analysis is exploratory.

3.2 Inputs and outputs in the eight conditions

Common inputs. Every request supplied an expert-role description, the SE setting, child and parent definitions, state labels and order, and assumed influence directions (whether higher parent states tend to increase or decrease the child). It listed the parent-state combinations to evaluate. Each answer conditioned on all listed parent states, averaging over unspecified circumstances. The persona supplied a professional perspective without claiming company observations. Graphs, definitions, prompts, and numerical procedures were fixed before collection.

Elicitation method. Direct requests listed every CPT row and asked for a probability vector over the child's states for each row.

Table 1: LLM-assisted BN construction compared by inputs, evaluation, and scope. Entries distinguish numerical CPT agreement from modeling support and downstream performance; reported errors are not directly comparable. KL denotes Kullback–Leibler divergence.

Study	Construction and supplied evidence	Evaluation material and endpoint	Scope relative to our comparison
Nafar et al. [18] TMLR, 2026	LLM internal knowledge; one conditional row or one child-state probability per query. A separate stage combines estimates with observations.	80 public BNs, eight LLMs: BN/CPT KL. Separate classification evaluation on nine datasets with three LLMs.	Changes granularity within a row; our RNM approach requests selected rows and reconstructs the rest.
Gottal and Matthes [12] Findings ACL, 2026	Reconstruction algorithms and direct LLM CPT output from numerical statistical summaries; separate extraction from publications.	Synthetic BNs and publication/raw-data cases; CPT distances, inference and decision utility, and runtime.	Quantitative summaries constrain construction; our RAG packets supply qualitative context.
Rahul and Chowdhury [22] Preprint, 2026	Persona-conditioned LLM panels provide distributions for parent configurations; trimmed-mean aggregation.	One healthcare-consultation BN; plausibility and internal consistency checks; calls, token estimates, and runtime.	Demonstrates construction feasibility and effort; lacks an independent numerical CPT benchmark.
Feng et al. (BIRD) [9] ICLR, 2025	LLM-generated factors and coarse likelihood judgments feed constrained parameter fitting and BN inference.	Human judgments of relative support and downstream task accuracy.	Establishes a precedent for LLM judgments plus fitting; does not test numerical reference-CPT recovery.
Cypko et al. [6] IJCARS, 2026	Clinical RAG supports expert BN modeling using guidelines and textbooks.	Synthetic retrieval evaluation; modeling, usability, and workload study with four clinicians.	Tests modeling support rather than complete numerical CPT agreement.
This study	Complete CPT elicitation versus LLM anchors plus RNM fitting; baseline/structured prompts, with/without qualitative RAG.	61 ordinal conditional nodes from seven SE BNs; complete-table distances, five replications, tokens, and fitting time.	Compares final CPTs on all rows of all nodes; checks weighting sensitivity and uses anchor fitting as a diagnostic.

Table 2: Evaluated BNs and CPTs. Rows count conditional distributions across the selected nodes; anchors are the subset requested in each RNM condition. Counts are per replication and prompt/evidence combination.

BN and source	Nodes	CPT rows	Anchors
Scrum [19]	25	249	123
Chrome 40 [14]	15	78	57
Chrome 42 [14]	12	63	46
APS [13]	3	8	8
SESSI [16]	1	3	3
Software logs 3 [8]	4	16	12
Software logs 4 [8]	1	4	3
Total	61	421	252

RNM requests listed only selected *anchors* and asked for their probability vectors plus a parent-combination rule: weighted mean, weighted minimum, weighted maximum, or minimum–maximum mixture. The LLM did not supply RNM weights or variance; an offline fitter estimated them from its answers. Both methods returned JSON with normalized vectors and a brief evidence note. Both received the same influence assumptions, but RNM additionally enforced them mathematically.

Prompt and evidence conditions. Each method used all four combinations below, producing eight conditions in total:

- **Baseline, no RAG:** common inputs and the method-specific output request.
- **Structured, no RAG:** additionally clarified conditioning, framed probabilities as expected counts among 100 hypothetical comparable cases, distinguished evidence from assumptions, and requested consistency checks.
- **Baseline, RAG:** baseline instructions plus relevant qualitative evidence cards.
- **Structured, RAG:** structured instructions plus the same evidence cards.

Structured prompting and RAG changed the support supplied to the LLM, not the requested rows or output format. Hypothetical counts aided judgment rather than supplying observations; fractional counts were allowed.

RAG cards paraphrased domain explanations. For example, a Scrum Cohesion card described shared goals and joint responsibility for tasks. This provides context without probabilities for Low, Medium, or High cohesion. Retrieval matched cards to model context and target/parent tags, selecting up to three; packets contained 24–103 words (median 72). Each node's packet was reused across methods, prompt styles, and replications. Cards excluded reference probabilities, numerical parameters, and identifying artifact links. RAG therefore tested additional qualitative context rather than live retrieval or company measurements. Input token budgets were not equalized.

3.3 Producing comparable complete tables

Direct outputs already contained complete CPTs. For RNM, the fitter used the LLM-selected rule and anchor distributions to estimate weights and variance, then generated all rows [11, 15]. It minimized mean anchor TV without accessing reference probabilities; valid poor fits were retained. Equations, numerical settings, and influence-direction evidence appear in the supplement.

Anchors comprised all-low, all-high, all-middle, one-high/rest-low, and one-low/rest-high configurations, with duplicates removed independently of the reference. For two ternary parents, direct elicitation requests nine distributions. RNM requests five: $(0,0)$, $(2,2)$, $(1,1)$, $(2,0)$, and $(0,2)$, where 0, 1, and 2 denote ordered states. Fitting produces a nine-row CPT, including fitted anchor rows. We compare all nine rows from each method with the same reference. Thus the comparison evaluates the complete construction pipelines, not only RNM's ability to fill unelicited rows.

3.4 Execution and reference isolation

Five replications were collected on 15–16 October 2026 using Codex CLI 0.162.1, requesting `gpt-6-astra` at `medium` effort. Each target–condition–replication used a fresh session; requests were shuffled within replications. Temperature and generation seed were unspecified. Experimental sessions could not access references, full papers, evaluator code, or authoring conversations; tools, browsing, connectors, and memory were disabled.

Outputs required correct rows/states, finite probabilities in $[0,1]$, and sums within 10^{-6} of one. One fresh-session format repair was allowed without reference feedback. All 2,440 trials produced valid outputs after four repairs (two per method); accounting includes all 2,444 attempts, with no fallback distributions.

3.5 Outcomes and analysis

For reference row p and prediction q, lower total variation means closer agreement:

$$\mathrm{TV}(p,q) = \frac{1}{2}\sum_{s}|p_s - q_s| \in [0,1]. \tag{1}$$

For each node, we average TV over all its rows. The primary full-sample score gives each of the 61 nodes equal weight; it therefore prevents larger CPTs from dominating solely through their row counts. RQ1 compares RNM minus direct after equally averaging the four prompt/evidence combinations within each replication. RQ2 contrasts structured minus baseline and RAG minus no RAG within each method, equally averaging the other factor. All eight condition means use the same complete-table endpoint.

The sample is unevenly represented: Scrum and the two Chrome BNs supply 52 of 61 nodes, so they dominate the node-weighted mean. To check this influence, we group BNs from the same publication into a family and give the five families equal weight, averaging nodes within BNs and BNs within families. This retains all nodes while reducing the influence of studies contributing more nodes. We report all seven BNs descriptively to expose variation behind either aggregate. Jensen–Shannon divergence, ordinal-CDF distance, and smoothed CPT KL use the same complete-table weights in supplementary results.

RQ3 counts requested distributions before repairs and client-reported tokens over all attempts for the full sample. Total tokens equal input plus output; cached-input and reasoning-output counters are subsets, not additional tokens. Paired contrasts use replication totals. Latency and offline fitting time are descriptive; subscription tokens are not converted into monetary savings.

The five paired replication blocks are the inferential units, not correlated nodes or rows. Two-sided paired t-tests ($df = 4$) and pointwise 95% intervals assume approximately normal, independent, stationary block differences. Five blocks provide little evidence for these assumptions. Two-sided exact sign tests assess direction/median sensitivity under independence; five concordant nonzero differences still yield $p = 0.0625$. Holm adjustments cover two RQ1 weighting contrasts, eight RQ2 contrasts (two methods, two factors, two weighting rules), and two RQ3 token contrasts. These are separate exploratory families, not study-wide confirmatory control; intervals are not simultaneous. No prespecified practical-effect or equivalence margin supports claims of usefulness or equivalence.

3.6 Offline diagnostics

Supplementary post-hoc diagnostics compare raw/fitted anchors, uniform predictions, alternative families, direct-anchor fits, stronger searches, and wider bounds. Reference-informed fits assess best-found approximation, not held-out accuracy or error floors. Non-anchor, Chrome-version, and node-omission diagnostics concern a 25-node compression subset. Their rows and weights are specified separately; they neither define the main endpoint nor replace its predictions.

4 Results

RNM requests fewer distributions but increases mean complete-table disagreement. All main results include every row of all 61 nodes; lower TV is better, and negative RNM-minus-direct differences favor RNM.

4.1 RQ1: Which method produces closer complete CPTs?

Direct elicitation has lower mean error across the sample. Mean TV is 0.2317 for direct elicitation and 0.2404 for RNM, an absolute increase of 0.0087 or 3.76% with RNM. The paired difference is positive in all five replications, ranging from 0.0072 to 0.0094 (Figure 1b). Its pointwise 95% t interval is $[0.0076, 0.0098]$, with Holm-adjusted $p = 0.000048$. The corresponding exact sign test does not reject ($p = 0.125$ after adjustment). Observed differences are similar across the five runs, but these blocks provide limited distribution-free evidence. Neither test establishes whether the difference matters for a particular SE decision.

Because Scrum and Chrome supply 52 of 61 nodes, we checked whether their numerical dominance explains the result. Giving each of the five source studies equal weight still favors direct elicitation: TV is 0.1937 for direct elicitation and 0.2189 for RNM, a difference of 0.0252 or 12.99%. Its interval is $[0.0211, 0.0292]$, adjusted t $p = 0.000065$, and adjusted sign $p = 0.125$. This is the same 61-node sample under different weights. The direction therefore persists

Table 3: Complete-table TV by BN, averaged over prompt/evidence conditions and replications. Δ is RNM minus direct. Every node is included; BN rows are descriptive.

BN	Nodes	Direct	RNM	Δ
Scrum	25	0.2605	0.2334	−0.0272
Chrome 40	15	0.2069	0.2220	+0.0151
Chrome 42	12	0.1988	0.2335	+0.0347
APS	3	0.0964	0.1155	+0.0191
SESSI	1	0.1160	0.1143	−0.0017
Software logs 3	4	0.3802	0.4770	+0.0967
Software logs 4	1	0.2052	0.3299	+0.1247
All nodes	61	0.2317	0.2404	+0.0087

when studies contributing more nodes receive less influence. Secondary complete-table distances also favor direct elicitation under both weighting rules (supplement).

The aggregate reflects unequal gains and losses. Table 3 and Figure 1a show every BN. RNM improves mean agreement for Scrum and slightly for SESSI, while both Chrome versions, APS, and both software-log BNs worsen. At node level, RNM is closer for 31 of 61 nodes and direct elicitation for 30. The overall mean favors direct elicitation because the magnitudes of deterioration outweigh the improvements; it is not a count of node-level wins. These descriptive comparisons reveal variation rather than establishing seven separate population effects.

Fitting can improve one relationship and weaken another. Consider a Scrum team with low self-organization but medium collaboration. Its Cohesion node represents working synchronously toward shared goals. For this configuration, the reference assigns 76.6% probability to Low cohesion. Direct elicitation assigns 58.0%, while RNM reconstructs 67.6%, moving closer to the reference (Figure 2a). Mean row TV falls from 0.1862 to 0.0901. Here, a relationship fitted to other elicited configurations improves a row of the final CPT.

Now consider a Chrome component with three or four contributors and no minor contributors. Contributors are classified by their share of changes. With no minor contributors, all must be major, so the Major Contributors node belongs entirely in the three-to-four category. Both the reference and all direct responses assign probability one to that category. RNM assigns 84.4%, spreading the remaining 15.6% to neighboring categories (Figure 2b). Smoothing through the fitted family weakens a count constraint preserved by direct elicitation. These deliberately contrasting examples explain possible changes, not their prevalence across all seven BNs.

Supplementary diagnostics compare raw and fitted judgments on identical anchors in the 25-node compression subset. Fitting changes Scrum anchor TV little (0.2790 to 0.2762) but raises Chrome anchor TV (0.2491 to 0.2994). Alternative families and searches, reference-informed approximation, and node-omission checks help examine that contrast, with their restricted scope stated separately. They do not establish a single cause of the full-sample result. **RQ1 does not support an overall RNM advantage: direct elicitation has lower mean complete-table TV under both weighting rules, despite improvements in some nodes and BNs.**

Table 4: All eight conditions: complete-table TV across all 61 nodes, with equal node weights and five replications. Each row pairs the methods under the same prompt and evidence.

Prompt	Evidence	Direct	RNM	Δ
Baseline	None	0.2327	0.2396	+0.0070
Baseline	RAG	0.2349	0.2419	+0.0070
Structured	None	0.2282	0.2389	+0.0107
Structured	RAG	0.2311	0.2413	+0.0102

4.2 RQ2: Do richer prompts or qualitative evidence help?

Direct elicitation has lower mean TV in every matched prompt/evidence combination (Table 4). Structured prompting lowers the direct method's node-weighted TV by 0.00413, with interval [−0.00595, −0.00232] and Holm-adjusted t $p = 0.0255$. This is the only RQ2 contrast that rejects after the eight-test correction. Under equal-family weighting, the direct prompt effect is −0.00508, but its interval crosses zero and adjusted $p = 0.4953$. Thus, the point estimates agree in direction while uncertainty differs with weighting. No RQ2 exact sign test rejects after adjustment.

For RNM, the node-weighted structured-prompt effect is −0.00067. Adding RAG changes TV by +0.00252 for direct elicitation and +0.00230 for RNM; none of these adjusted tests rejects. The Cohesion packet, for example, describes shared goals without stating how often cohesion is Low given Low self-organization. It adds context to definitions rather than numerical constraints. **RQ2 provides limited support for structured direct elicitation, but no demonstrated benefit from the tested qualitative RAG.** This does not establish equivalence or the ineffectiveness of local quantitative evidence. Full contrasts under both weights appear in the supplement.

4.3 RQ3: Do fewer judgments translate into fewer tokens?

RNM requests 5,040 distributions across the experiment, compared with 8,420 for direct elicitation: 40.1% fewer. Both methods still make one initial request per node and condition. Reducing the returned table therefore does not eliminate the shared instructions, definitions, and evidence sent with each request.

Output tokens fall by 12.12%, but input-plus-output tokens fall by only 0.61% (Table 5). Inputs dominate: approximately 14.36 million for direct elicitation and 14.31 million for RNM. Across five paired replication totals, output savings have adjusted t $p = 0.000073$, whereas total-token savings do not reject ($p = 0.0809$). Both adjusted exact sign tests give $p = 0.125$. The supplement reports block differences and intervals. Fewer requested probabilities therefore do not yield proportional computational savings.

Summed request latency is 3.82% lower for RNM; the original retained fits sum to 182.3 seconds of recorded fitting time, excluding subsequent offline diagnostics. These are computational descriptions, not controlled wall-clock or monetary savings. **RQ3 supports reduced elicitation and output size, with limited evidence of total-token savings.** Those savings accompany higher

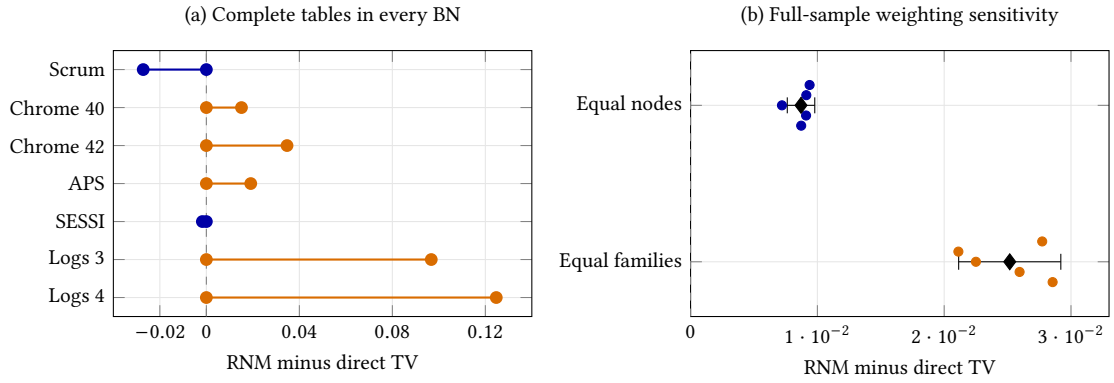


Figure 1: Complete-CPT agreement across the full sample. Negative differences favor RNM. (a) Each BN's mean averages all rows within nodes, then nodes, four prompt/evidence combinations, and five replications equally; points are descriptive. (b) Both weighting rules retain all 61 nodes. Circles show the five block differences; diamonds and whiskers show means and pointwise 95% t intervals. Equal-family weighting first averages nodes within BNs and BNs within each of five publication families.

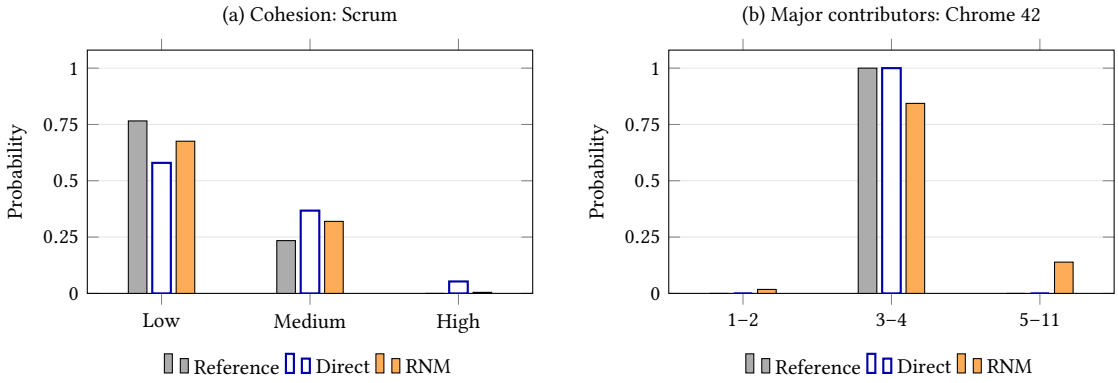


Figure 2: Two CPT rows unqueried by RNM. (a) Cohesion given Low Self-Organizing and Medium Collaboration. (b) Major contributors given three or four total contributors and no minor contributors. Bars average 20 predictions per method. Examples were selected for opposing effects in the earlier non-anchor comparison; they illustrate mechanisms and do not represent the full sample.

Table 5: RQ3: tokens across all 61 nodes and five replications, including repairs. Holm-adjusted paired-t tests compare five method-paired block totals across two counters; neither exact sign test rejects.

Counter	Direct	RNM	Reduction	Holm t p
Output	341,153	299,798	12.12%	0.000073
Input + output	14,697,090	14,607,517	0.61%	0.080921

mean complete-table error across the sample, so they do not establish a superior accuracy–effort tradeoff.

5 Discussion

The full-sample comparison exposes an accuracy–effort tradeoff: RNM requests fewer judgments and produces shorter answers, yet its final CPTs have higher mean reference error. Network-level results show where that summary applies and where it does not. We interpret the three RQs against prior work, then consider SE research and practice.

5.1 Research questions in relation to the literature

RQ1: Reference agreement. RNM's partial-elicitation rationale [11, 15] does not ensure that fitting LLM judgments improves final CPTs. Direct elicitation has lower mean error under both full-sample

weighting rules, but RNM is closer on a small majority of nodes. The magnitude and distribution of errors therefore matter more than a count of wins. Equal-family weighting increases RNM's disadvantage because the smaller software-log BNs have large deteriorations, while the 25-node Scrum BN improves. Neither weighting estimates a universal effect across SE organizations.

Compatibility between the reconstruction and the intended relationships is a plausible explanation. Scrum's reference was itself RNM-derived; the contributor-count example shows how fitted smoothing can weaken a constraint. Restricted anchor and search diagnostics support this explanation for the Scrum–Chrome contrast, without proving its cause or explaining every BN. Reference construction, influence assumptions, elicited judgments, and optimization can all contribute. Blomaard et al.'s comparison optimizes against known CPTs [2], while our experimental fits use only LLM judgments. Those designs address different limits of partial elicitation. The seven-BN results caution against inferring numerical fidelity from ordinal eligibility alone.

RQ2: Elicitation support. Structured prompting modestly improves node-weighted direct elicitation, but adjusted significance does not persist under equal-family weighting. It supplies no demonstrated benefit for RNM. Nafar et al. distinguish changes in probability-query granularity from prompt modifications [18]; their within-row elicitation gains do not establish an advantage for omitting rows and reconstructing tables. Our qualitative RAG adds no demonstrated improvement beyond shared definitions. Clinical modeling support [6] and CPT reconstruction from statistical summaries [12] involve different evidence and endpoints. Local measurements may constrain probabilities in ways that our qualitative cards cannot.

RQ3: Accuracy and effort. The human burden–fidelity tradeoff motivates partial elicitation [1], but fewer requested judgments do not translate proportionally into LLM token savings. Shared inputs dominate this implementation's workload. The observed 12.1% output reduction accompanies only 0.61% fewer total tokens and higher mean CPT error. This is evidence about the tested request design, not a claim that all RNM implementations have the same cost profile.

5.2 Implications for SE researchers and practitioners

Researchers should evaluate the complete constructed artifact, retain every eligible case, and show variation behind aggregate scores. Stating the weighting rule makes clear whether a result describes the sampled nodes or equally weighted application families. Anchor comparisons can diagnose fitting changes, but should complement final-table evaluation. Work combining neural judgments and structured inference, such as BIRD [9], also motivates evaluating downstream predictions and decisions instead of assuming that inspectability guarantees usefulness.

For practitioners, direct elicitation is the stronger average numerical baseline in this sample; RNM remains a candidate when its assumptions fit the intended relationships or output burden is the priority. Neither method is validated for deployment by CPT agreement alone. Check meanings, influence directions, constraints, and probabilities against organizational measurements or experts. A small RNM fitting residual shows agreement with the elicited judgments, not with the organization. Reusable BN components may support initialization, but tailoring and held-out organizational validation remain necessary before decision use.

6 Threats to Validity

Construct validity. TV measures contextual CPT agreement, not observed calibration, expert replacement, explanation quality, or decision utility. Equal-row scoring includes potentially rare configurations. Scrum's reconstruction lacks an independently certified AgenaRisk export, and its RNM origin favors the tested representation. The seven-BN breakdown and restricted fitting diagnostics expose this distinction. SESSI represents an original elicitation stage. Adapted variable meanings and uncertain parent directions may affect agreement despite correct indexing.

Internal validity. Fresh sessions, shared definitions, reference isolation, and blinded fitting limit direct access leakage. Public benchmark memorization remains possible. The logs identify the requested model configuration, but cannot independently verify the exact serving-model version. Prompt and fitting development used the Scrum setting, which may favor that reference. Offline variants cannot exhaust all explanations. Reference-informed selection is optimistic in-sample diagnosis; bounded search may miss better fits. Schema validation cannot establish semantic or evidence fidelity, and exhaustive human review was not performed. The elicitation instructions and evidence cards were tested as bundles, so the contribution of each individual instruction or passage is unknown.

Conclusion validity. Five blocks give little evidence about the distribution or stationarity of differences. The paired-t and exact-sign results must be considered together; neither treats shared nodes or rows as independent replications. The SE restriction, completetable primary endpoint, weighting choices, and statistical plan were selected after inspecting earlier results; inference is therefore exploratory. Separate Holm families do not provide study-wide confirmatory error control. No prespecified acceptable-error margin supports claims of equivalence or practical sufficiency. Repeated execution estimates here are conditional on these fixed references and settings.

External validity. The 61-node sample spans seven BNs but only five publication families; Scrum and Chrome supply 52 nodes. Equal-family sensitivity changes their influence without making the sample representative. Some nodes offer no reduction in elicited rows, and tables have at most 27 rows. Only one requested model and reasoning effort were evaluated. Organizational distributions, larger CPTs, other model families, and evidence containing local numerical measurements may yield different tradeoffs.

7 Conclusion

We compared direct LLM elicitation with LLM-assisted RNM on complete CPTs for 61 ordinal nodes from seven SE BNs, including anchors. Eight conditions varied method, prompting, and qualitative RAG over five replications.

Direct elicitation achieved lower mean TV: 0.2317 versus 0.2404, a 3.76% increase with RNM. Equal publication-family weighting retained the direction and increased the difference. RNM nevertheless

improved 31 of 61 nodes and the mean for Scrum and SESSI; larger deteriorations elsewhere outweighed those gains. Reconstruction therefore cannot be recommended from ordinal eligibility or reduced input size alone. Both overall weighting contrasts passed Holm-adjusted paired t-tests, but not exact sign tests with five replications, limiting the strength of the statistical evidence.

Structured prompting provided a modest benefit for node-weighted direct elicitation, without consistent statistical support across weighting rules. Qualitative RAG demonstrated no benefit. RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, yet only 0.61% fewer total tokens. Its savings in returned information thus accompanied higher mean complete-table error, rather than establishing a superior accuracy–effort tradeoff.

Researchers should report complete-pipeline comparisons, weighting sensitivity, and network variation. For practitioners, LLM-generated CPTs remain candidate initializations requiring validation of probabilities and constraints in the intended organization. Future work should test local quantitative evidence, additional model configurations, larger CPTs, and prediction or decision quality on held-out projects. This would connect reference agreement to explainable SE decision support.

Replication Materials

The replication package contains the elicitation materials, reference CPTs, saved responses, fitted tables, evaluation scripts, token counts, and supplementary analyses. Reference materials are separated from collection inputs. The repository is private for author review.

Use of Generative AI

LLM judgments are experimental data (Section 3). Generative AI assisted drafting and script preparation/checking; numerical claims trace to saved responses and deterministic analyses.

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